

SMU Beamer Theme

Your Name

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01

Basic Settings

The SMU Beamer Theme provides a LaTeX beamer template based on SMU's official PowerPoint template, with enhanced features for academic presentations:

- Official SMU branding and colors
- Custom block components
- Flexible layout options

Design & Layout:

- Professional SMU branding
- Progress bar footer
- Section transition pages
- Multi-column layouts

Text & Colors:

- Custom color commands
- Text highlighting
- Bold color variants

Custom Blocks:

- Standard blocks
- Alert blocks
- Mathematical boxes (Theorem, etc.)
- Tagged blocks

Font Styles:

- Regular text in Calibri font
- **Bold text appears in SMU blue**
- *Italic text for emphasis*
- **Important warning** using red bold
- **Key concept** using blue bold

Inline Code Style:

- Use `inline` code for technical terms
- Function names like `printf()` or `main()`
- Styles can be customized in `smucore/smucode.def`

SMU Brand Colors:

- SMU Blue - Primary color
- SMU Light Blue - Secondary color
- SMU Gold - Accent color
- SMU Green - Additional accent

Additional Accent Colors:

- SCIS Yellow - School of Computing
- Alert Red - For warnings
- Muted Blue - Alternative blue
- Muted Green - Alternative green

Inline Mathematics:

Einstein's famous equation $E = mc^2$, Euler's number $e \approx 2.71828$, and pi $\pi \approx 3.14159$.

Display Equations:

$$\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Aligned Equations:

$$\begin{aligned} f(x) &= x^2 + 2x + 1 \\ &= (x + 1)^2 \\ &= (x + 1)(x + 1) \end{aligned}$$

Basic Highlighting:

Normal text with default highlighted content using SMU blue.

Custom Colors:

Different colors: red highlight, green highlight, yellow highlight, and blue highlight.

Math Formula Highlighting:

Inline math: $E = mc^2$ and $a^2 + b^2 = c^2$

Display math:
$$\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

02

Components Design

Standard

This is a standard block with SMU styling.

Alert

Warning block for important information.

Examples

Example block for demonstrating concepts.

Light box for highlighting supplementary information.

Tag

Tagged block for categorized content.

Theorem: Pythagorean Theorem

For a right triangle with sides a and b , and hypotenuse c :

$$a^2 + b^2 = c^2$$

The `\begin{twocolumns}` command defaults to 50% width for each column. You can use `\leftcol[0.6]` and `\rightcol[0.35]` to customize the column widths.

Wider Left Column (60%)

This column takes 60% of the available width, leaving 35% for the right column (plus some spacing).

Useful for:

- Main content vs. sidebar
- Text vs. small figure
- Detailed explanation vs. summary

Narrow Right (35%)

Smaller column for supplementary information.

Step 1

Initialize

Input: Data

Step 2

Process

Transform: Apply

Step 3

Output

Result: Done

Three columns are perfect for showing processes or comparisons.

Example: Performance Comparison

Method	Accuracy (%)	Time (ms)	Memory (MB)
Method A	95.3	12.5	128
Method B	97.1	18.2	256
Method C	96.8	15.7	192
Ours	98.5	14.3	160

Table: Performance comparison of different methods

Use booktabs package for professional-looking tables.

Unordered Lists (Itemize):

- First level item
- Another first level item
 - Second level nested item
 - Another second level item
 - Third level deeply nested
 - More third level content
 - Back to second level
- Back to first level
- Final first level item

Ordered Lists (Enumerate):

1. First step in the process
2. Second step with details
 - 2.1 Sub-step 2.1
 - 2.2 Sub-step 2.2
 - 2.2.1 Sub-sub-step 2.2.1
 - 2.2.2 Sub-sub-step 2.2.2
3. Third step returning to main level
4. Fourth and final step

Custom Description Lists:

- API** Application Programming Interface for software communication
- SDK** Software Development Kit containing tools and libraries
- IDE** Integrated Development Environment for coding
- CI/CD** Continuous Integration and Continuous Deployment

Description lists are perfect for glossaries or term definitions.

Getting Started

Load the theme: `\usepackage{smu}`

Columns

Two columns: `\begin{twocolumns}...\end{twocolumns}`

Three columns: `\begin{threecolumns}...\end{threecolumns}`

Documentation

See `README.md` for detailed documentation and examples.

The theme is highly customizable:

Colors Edit `smucore/smucolors.def`

Fonts Edit `smucore/smufonts.def`

Commands Edit `smucore/smucommands.def`

Blocks Edit `smucore/smublocks.def`

Templates Edit `smucore/smutemplates.def`

All components are modular and can be customized independently!

Documentation

- README.md - Main documentation
- main.tex - This demo file
- Source files in smucore/ directory

External Resources

- Beamer: <https://ctan.org/pkg/beamer>

MtA-based Approaches (Multiplicative-to-Additive)

- Paillier Encryption: Requires strong-RSA assumption.
[GGN+16; Lin+17; GG+18; CGG+20; XLX+24]
- CL Encryption: Based on Class Groups.
[CCL+19; CCL+20; YCX+21; DMZ+21]
- Oblivious Transfer: No extra assumptions and computationally efficient.
[DKL+18; DKL+19; DKL+24]
- Joye-Libert: Based on revisited Joye-Libert encryption.
[XAL+23]

Threshold CL Encryption

- Decryption key of CL shared among n parties.
[BDO+23; WMY+23; WMC+24; TX+25; JTX+25]

Others

- Pseudorandom Correlation Generator [ANO+22], ...

- [ANO+22] Damiano Abram, Ariel Nof, Claudio Orlandi, Peter Scholl, and Omer Shlomovits, Low-bandwidth threshold ECDSA via pseudorandom correlation generators. In: *2022 IEEE Symposium on Security and Privacy (SP)*, IEEE, 2022, pp. 2554–2572.
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- [GG+18] Rosario Gennaro and Steven Goldfeder, Fast multiparty threshold ECDSA with fast trustless setup. In: *Proceedings of the 2018 ACM SIGSAC conference on computer and communications security*, 2018, pp. 1179–1194.
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- [TX+25] Guofeng Tang and Haiyang Xue, Robust Threshold ECDSA with Online-Friendly Design in Three Rounds. In: *2025 IEEE Symposium on Security and Privacy (SP)*, IEEE, 2025, pp. 203–221.
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Thank You

Q & A